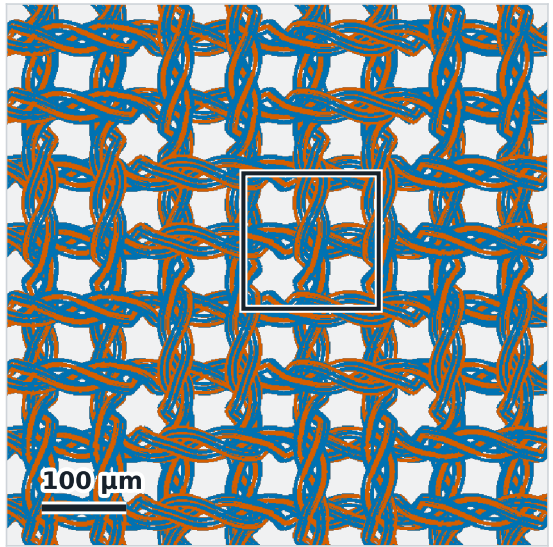
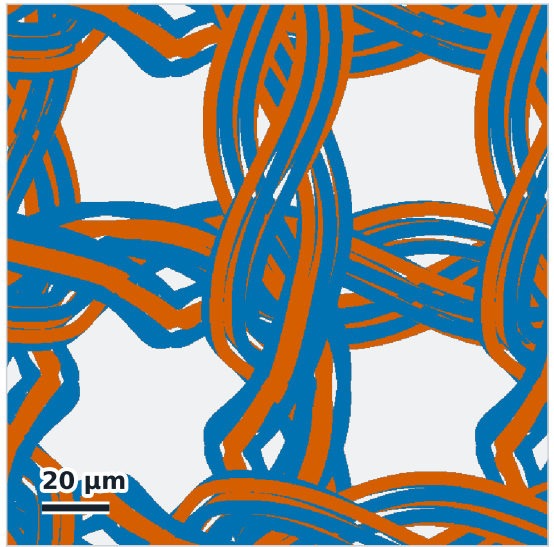
Fiber identity and local geometry

Integer material assignments stay attached to individual fibers across scales

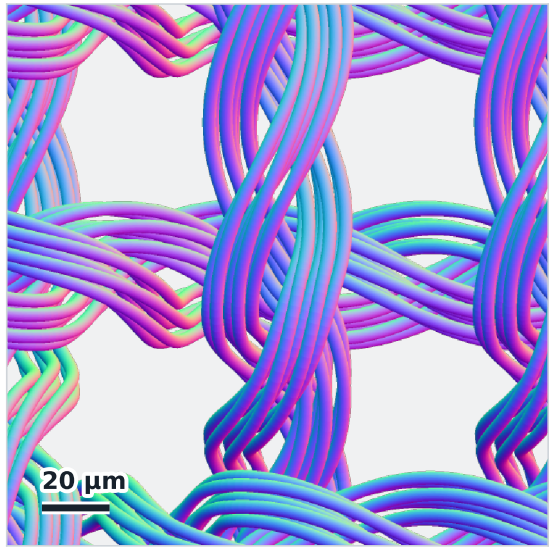
(a) Fiber material IDs



(b) Crossing detail: IDs



(c) Crossing detail: normals



synthetic_A synthetic_B

Material colors are categorical labels; they are not dye colors or inferred fiber chemistry.

IDs use pixel-center samples without category averaging · normal colors use real supersamples · light gray = no surface hit